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The Editors
Cell Systems, Cell Press

Dear Editors,

I am pleased to submit my manuscript, “Sound Biosynthetic Design over Iterative Grammars: A Realizability Geometry for Fungal Polyketides,” for consideration as a Research Article in *Cell Systems*.

Predicting a fungal natural product from its biosynthetic gene cluster has stalled at 51–68% balanced accuracy, and pooling a thousand bacterial clusters with the few hundred fungal ones barely moves that ceiling. This manuscript argues the target is mis-specified. A product is the image of a latent biosynthetic program executed under a domain-derived grammar, so the ceiling is grammar *identifiability*—how much of the program the observables pin down—not data volume. Recast this way, prediction and design become one quantity: the programs consistent with an observation and the programs an engineer may specify toward a target are the same observability class. The manuscript proves this as an identifiability–controllability duality, so a single engine serves both directions.

Concretely, the work builds a sound, grammar-agnostic inverse compiler. A deterministic executor renders a program to a structure; a beam verifier returns the exact consistent program set as a typed verdict—*verified*, *under-observed*, or *out-of-grammar*—where the under-observed state names the next observable that would collapse the ambiguity. Run in inference over the 326 fungal PKS clusters in MIBiG 4.0, exact reconstruction reaches nine cores; a corpus-wide census establishes this as a genuine boundary of the current sound grammar rather than a low score, and sorts the remainder onto a located expressibility ladder. In the design direction, the same engine places a target on a realizability geometry scored by a literature-curated cost that encodes the field’s measured engineering difficulty, and a planner ranks the experiment that most collapses the candidate set per unit cost. The observability and verdict layers are family-agnostic, demonstrated unchanged across PKS, a non-ribosomal peptide (NRPS) grammar, and a typed PKS–NRPS hybrid.

The work suits the quantitative, computational readership of *Cell Systems*: its contribution is a systems-level principle—an observability quantity that unifies prediction and design—instantiated in an open, reproducible tool and grounded in a sound verifier rather than a single accuracy figure. I am explicit about scope: the engine is a metabologenomics triage tool that names the clusters where accurate mass and tandem mass spectrometry are insufficient and structure elucidation is required, not a replacement for elucidation.

The manuscript is original, has not been published, and is not under consideration elsewhere. All code, data, and analyses are openly available, and every reported number reproduces from a single documented command (<https://github.com/BrageEilertsen/lpi-fungal>). I am the sole author and declare no competing interests. I would gladly suggest reviewers on request.

Thank you for considering this work.

Sincerely,

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